

Zachary M. Deak, DO

CONTACT INFORMATION

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RESEARCH INTERESTS

Wearable biosensing systems for continuous health monitoring in operationally demanding and resource-constrained environments, such as long-duration spaceflight. Particular interest in applying novel sensing chemistries and transduction materials, minimally-invasive intradermal platforms, and the end-to-end translation of molecular diagnostics into deployable wearable devices.

Keywords: continuous biomarker monitoring, microneedle architectures, electrochemical and optical transduction, chemometrics, spaceflight physiology, translational medicine.

EDUCATION

Georgia Institute of Technology, Atlanta, GA 2026–present

PhD, Human-Centered Computing, School of Interactive Computing

Fellowship: NSF NRT, Sustainable Development of Smart Medical Devices (SUSMED)

Advisor: Alexander T. Adams, PhD (Uncommon Sense Lab)

Michigan State University, East Lansing, MI 2018–2022

DO, Doctor of Osteopathic Medicine, College of Osteopathic Medicine

COMLEX-USA Level 1: 90th Percentile

Clerkships: Detroit Medical Center

Central Michigan University, Mount Pleasant, MI 2014–2018

BS, Biomedical Sciences, Department of Biology

HONORS & AWARDS

Research Fellowship, NSF NRT, Sustainable Development of Smart Medical Devices 2026
Georgia Institute of Technology

NASA Young Scientist Program Graduate 2026
Blue Marble Space Institute of Science

First Quintile Graduate, Highest Academic Distinction 2022
Michigan State University College of Osteopathic Medicine

Sigma Sigma Phi Service Award 2022
Michigan State University College of Osteopathic Medicine

Summa cum laude, 3.92 GPA 2018
Central Michigan University

Honors Program Graduate 2018
Central Michigan University

Undergraduate Research and Creative Endeavors Grant, \$1000 2017
Central Michigan University

RESEARCH EXPERIENCE

NASA Ames Research Center, Moffett Field, CA 2026
Biosensor Research & Development Intern • Jessica E. Koehne's Lab

- Developed an electrochemical aptamer-based (EAB) microneedle biosensor patch for continuous interstitial fluid cortisol monitoring to support crew health during long-duration spaceflight.
- Fabricated gold-coated microneedle arrays via SLA 3D printing and magnetron sputter deposition (Cr/Au); verified surface morphology via SEM and established an H_2SO_4 oxide-stripping sequence to restore electron-transfer kinetics.
- Characterized aptamer monolayer formation via dual-frequency square-wave voltammetry, optimizing thiolated-aptamer and MCH backfill concentrations for well-resolved methylene-blue signaling in simulated ISF.

- Identified an unusual presentation of lung adenocarcinoma with soft tissue metastases during a surgical clerkship and led development into a formal case report.
- Managed publication workflows including patient consent, biopsy coordination, pathology imaging, literature review, and manuscript preparation.
- Coauthored the resulting publication describing rare metastatic patterns of lung cancer.

- Designed and evaluated gradient transplantation of neurotrophin-3 (NT-3) overexpressing mesenchymal stem cell (MSC) grafts for motor recovery in a rat model of complete spinal cord transection.
- Performed cell culture, tissue preparation, immunohistochemistry, RT-PCR, flow cytometry, and animal behavioral testing.
- Contributed to experimental design, data analysis, and figure preparation for publication.

- Contributed to projects evaluating mesenchymal and neural stem cell transplantation strategies, including SDF-1 therapy, for spinal cord injury repair.
- Performed cell culture, post-mortem tissue analyses, immunohistochemistry, flow cytometry, and behavioral assessments.
- Assisted with manuscript development, figure preparation, and literature review.

- Investigated enzymatic mechanisms involving NAD and FAD electron carriers.
- Validated reaction products via ¹H and ¹³C NMR spectroscopy, mass and infrared spectroscopy.

PUBLICATIONS

- Tvedten E, **Deak Z**, Schwartz B, Rice A. An Atypical Presentation of Soft Tissue Metastasis in a Patient With Lung Cancer. *Cureus*. 2021;13(7):e16294.
- Stewart AN, Kendzierski G, **Deak ZM**, et al. Transplantation of Mesenchymal Stem Cells that Overexpress NT-3 Produce Motor Improvements without Axonal Regeneration following Complete Spinal Cord Transections in Rats. *Brain Research*. 2018;1699:19-33.
- Stewart AN, Kendzierski G, **Deak ZM**, et al. Co-Transplantation of Mesenchymal and Neural Stem Cells and Overexpressing Stromal-Derived-Factor-1 for Treating Spinal Cord Injury. *Brain Research*. 2017;1672:91-105.

PRESENTATIONS

- Deak ZM**, Cordeiro M, Anderson MJ, Koehne JE. 3D-printed aptamer-based microneedles as a platform for real-time continuous cortisol monitoring during spaceflight. Ames Summer Poster Session, NASA Ames Research Center, Moffett Field, CA. August 2026. Poster.
- Deak ZM**, Kendzierski G, Stewart AN, Dunbar GL, Rossignol J. Engineering Mesenchymal Stem Cells to Overexpress Neurotrophin-3 as a Treatment for Spinal Cord Injury. Capitol Scholars Exhibition, Michigan State Capitol, Lansing, MI. March 2018. Poster.
- Deak ZM**, Kendzierski G, Stewart AN, Dunbar GL, Rossignol J. Transplantation of Neurotrophin-3 Overexpressing Mesenchymal Stem Cells Rostral and Caudal to the Epicenter Following Spinal Cord Injury. CMU Student Creative Research and Endeavors Exhibition, Mount Pleasant, MI. April 2018. Poster.

TECHNICAL PROJECTS

3D Rendering Engine

2024–present

C99, Vulkan, Bash

- Developing a custom visualization engine for real-time rendering of anatomical models and medical imaging data for educational and diagnostic use.
- Implementing a DICOM file parser and viewer for interactive exploration of medical imaging datasets.

Medical Boards Prep Platform (MedStudForty)

2022–2026

TypeScript, React/Next.js, MySQL

- Built an interactive platform emulating USMLE and COMLEX examination formats to improve preclinical diagnostic reasoning.
- Developed a real-time timed testing interface with user authentication, session analytics, and SEO optimization.
- Designed and deployed a custom content management system and instructor dashboard for question authoring and performance tracking.

TEACHING EXPERIENCE

Founder and Educator, MedStudForty

2022–2026

Online Medical Education Platform

- Delivered USMLE and COMLEX style question banks and data-driven, individualized tutoring based on student performance analytics.
- Authored hundreds of board-style practice questions with evidence-based explanations aligned to NBME and NBOME blueprints.

Neuroanatomy Teaching Assistant (Bluecoat)

2020

OST 571, Neuromusculoskeletal System • Michigan State University College of Osteopathic Medicine

- Led small-group, prosection-based neuroanatomy labs for first-year medical students; facilitated discussion and proctored exams.

Human Anatomy Teaching Assistant (Bluecoat)

2019

ANTR 510, Clinical Human Gross Anatomy • Michigan State University College of Osteopathic Medicine

- Facilitated prosection-based gross anatomy labs emphasizing structure-function relationships and clinical application; ensured respectful handling of anatomical donors.

LEADERSHIP & SERVICE

Invited Alumni Speaker, Honors Program Alumni Series

2026

Central Michigan University

- Delivered an invited talk to undergraduate honors students on navigating graduate and medical school applications, interdisciplinary research, and career paths.

Medical Student Liaison, Michigan Osteopathic Association

2018–2022

Michigan State University College of Osteopathic Medicine

- Served as liaison between the Michigan Osteopathic Association and the MSUCOM student body, communicating policy updates and professional development opportunities.

SKILLS & TRAINING

Programming: C, C++, Python, TypeScript, Go, OCaml

Development: React, Next.js, Prisma, SQL, Bash, Git, Linux

Visualization & Rendering: Vulkan, OpenGL, Blender, DICOM

Fabrication & Prototyping: AutoCAD, CNC, laser engraving, 3D and SLA printing, Arduino

Electrochemistry: Aptamers, cyclic/square-wave voltammetry, potentiostat (Metrohm Autolab)

Molecular & Cellular Biology: Cell culture, immunohistochemistry (IHC), RT-PCR, flow cytometry

Materials & Spectroscopy: Scanning Electron Microscopy (SEM), mass spectrometry, FTIR spectroscopy